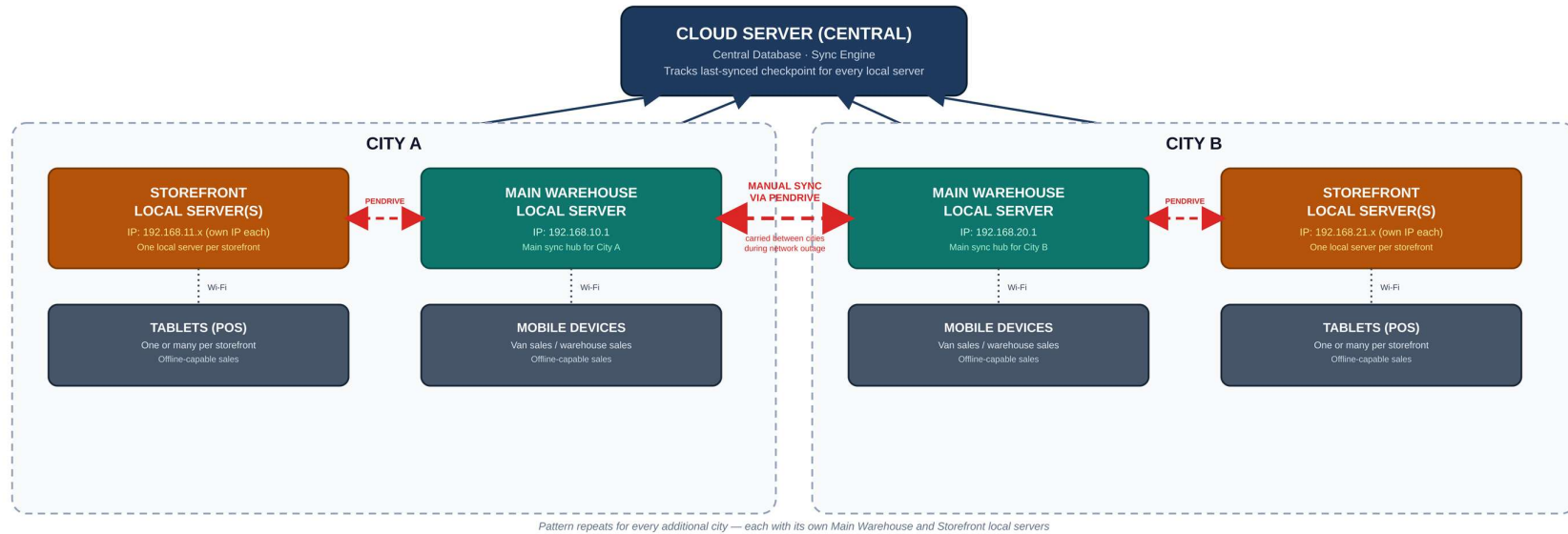


Offline-Resilient Local Server ↔ Cloud Sync Architecture

Multi-city POS system designed for regions with extended internet outages (e.g. Libya, 1–3 week disconnections)

Local Server ↔ Cloud Sync Architecture — Offline-Resilient Multi-City POS System



LEGEND

- Automatic sync over the internet — continuous, whenever the connection is available
- Manual sync via pendrive — physically carried, used during network outages
- Local Wi-Fi — device manually connected/paired to its own local server

How the Architecture Works

1. Normal Operation — Online

Every city has its own set of local servers: one Main Warehouse Local Server (the sync hub for that city) and one Local Server per Storefront, each with its own static IP address on the local network. Under normal conditions, all of these local servers stay continuously connected to the internet and auto-sync with the central Cloud Server, which acts as the single source of truth and tracks the last-synced checkpoint for every individual node.

2. Store & Warehouse Devices — Local Wi-Fi

- Storefronts: each storefront has one or more tablets running the POS app. Tablets are manually connected to that specific storefront's own local Wi-Fi network — they are paired to one storefront only.
- Warehouses: van sales and warehouse sales staff use mobile devices that operate offline. They are manually connected to the warehouse's local Wi-Fi and sync their offline sales data to the Warehouse Local Server whenever in range.

3. Network Outage — Manual Pendrive Sync

When the internet connection drops for an extended period (1–3 weeks is common in the target region), the Main Warehouse Local Server loses its link to the Cloud Server. Two manual sync paths keep data moving during this window:

- Storefront ↔ Warehouse (within a city): storefront data is periodically consolidated onto the city's Main Warehouse Local Server via a manually carried USB pendrive — the same mechanism used for the cross-city sync below.
- Warehouse ↔ Warehouse (across cities): a user copies the data from one city's Main Warehouse Local Server onto a pendrive, physically travels to another city, and uploads it into that city's Main Warehouse Local Server — bringing both warehouses up to date with each other despite having no internet.

4. Reconnection — Cloud Reconciliation

Once the internet connection is restored, each Main Warehouse Local Server resumes automatic sync with the Cloud Server. Because the cloud tracks the last-synced checkpoint per local server, it can determine exactly which records are new or changed at each node (including anything picked up through the manual pendrive transfers) and merge only the outstanding changes — avoiding duplication and keeping every city, warehouse, and storefront consistent.

Key Design Points

- Every local server (warehouse and storefront) has its own fixed IP address on its local network.
- Tablets and mobile devices connect to only one local server's Wi-Fi at a time — connections are set up manually, not roamed automatically.
- The pendrive transfer is the same mechanism whether it's storefront-to-warehouse or warehouse-to-warehouse — a manual, physical carry of an export/import package.
- The cloud is checkpoint-aware, so intermittent, out-of-order, or delayed syncs from any node still reconcile correctly once connectivity returns.